

Lecture 12 - Uncertainty quantification and sensitivity analysis

Monte Carlo, PCE, parameterizations

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1 Objectives

In this lecture, we will learn about how to model impact of uncertainty in the input parameters on the output of a simulation.

1. Types and source of uncertainty
2. Statistical analysis of uncertainty and sensitivity
3. Numerical methods for UQ/SA

2 Uncertainty

Where does it come from?

2.1 Sources of uncertainty in simulations

- Uncertainty in boundary conditions (free stream conditions, boundary values, surface properties, etc.)
- Uncertainty in model parameters (turbulence model constants, thermodynamic properties, etc.)

2.2 Classification of uncertainty

- Aleatoric: inherent, unavoidable randomness in the system (cosmic radiation, thermal noise, wind speed variations, etc.). **Not** due to lack of knowledge
- Epistemic: reducible uncertainty in model parameters, i.e. uncertainty **due** to lack of knowledge

2.3 Dealing with uncertainty

- Aleatoric: we can assess impact of the uncertainty on the simulation
- Epistemic: we can assess impact of the uncertainty on the simulation **and** potentially reduce uncertainty via better measurements, model inversion

2.4 Notation

Let Y be a scalar quantity of interest (QoI) that depends on N uncertain variables ξ :

$$Y = Y(\xi), \quad \xi = (\xi_1, \xi_2, \dots, \xi_N).$$

We assume that ξ are random variables with a known probability density function (PDF):

$$f(\xi; \mathbf{p}).$$

$\mathbf{p}$ is a set of parameters of the PDF. For example, for a Gaussian distribution we have the mean and standard deviation as parameters:

$$\mathbf{p} = (\mu, \sigma).$$

2.5 Inverse problems

In case of *epistemic* uncertainty, we might want to update parameters $\mathbf{p}$ based on measurements/additional data.

For Gaussian distributions ($\mathbf{p} = (\mu, \sigma)$), we might want to

- reduce σ (reduce uncertainty in ξ_i)
- adjust μ (update the model).

i Note

We drop any spatial and temporal dependence here for simplicity, the methods stay the same for varying quantities, as well as vector-valued QoIs.

2.6 Independently distributed variables

In general, we have that variables may be dependent on each other. In case of independent variables, the joint PDF is the product of the individual PDFs:

$$f(\xi; \mathbf{p}) = \prod_{i=1}^N f^{(i)}(\xi_i; \mathbf{p}_i).$$

From now on, the dependency on $\mathbf{p}$ ($\mathbf{p}_i$) is dropped for simplicity.

3 Statistical analysis of uncertainty and sensitivity

How do we quantify uncertainty and sensitivity?

3.1 Global vs local sensitivity

Question

If we know local dependency of Y on uncertain variables ξ_i , i.e. $\frac{\partial Y}{\partial \xi_i}$ (for example, via auto-differentiation), why do we need some statistical analysis?

Answer

Local dependency only shows how much Y changes if we change ξ_i by a small amount. It does not tell us how much Y changes due to statistical distribution of ξ_i !

3.2 Statistical evaluation

We can consider our QoI Y as randomly distributed quantity. We can define the mean value

$$\mathbb{E}[Y] = \int Y(\xi) f(\xi) d\xi$$

and variance

$$\text{Var}[Y] = \int (Y(\xi) - \mathbb{E}[Y])^2 f(\xi) d\xi.$$

Variance **is** a measure of uncertainty.

Note

For Lipschitz-continuous functions $Y(\xi)$ and independently distributed ξ_i , the variance can be bounded by the variance of the input + some distribution-specific constants.

3.3 Global sensitivity

Hoeffding decomposition / ANOVA (“Analysis of Variance”): in case of **independently distributed** ξ_i , we can write

$$Y(\xi) = Y_0 + \sum_{i=1}^N Y_i(\xi_i) + \sum_{1 \leq i < j \leq N} Y_{ij}(\xi_i, \xi_j) + \dots + Y_{1,\dots,N}(\xi_1, \dots, \xi_N).$$

Here

$$Y_0 = \mathbb{E}[Y], \quad Y_i = \mathbb{E}[Y|\xi_i] - Y_0, \quad Y_{ij} = \mathbb{E}[Y|\xi_i, \xi_j] - Y_0 - Y_i - Y_j, \dots$$

We can use this to decompose variance of Y !

3.4 Global sensitivity

Denote multi-index $\mathbf{u} \subset \{1, \dots, D\}$. Due to independent distribution of variables, we have

$$\text{Var}[Y] = \sum_{\mathbf{u}} \text{Var}[Y_{\mathbf{u}}].$$

Introduce Sobol indices:

$$S_{\mathbf{u}} = \frac{\text{Var}[Y_{\mathbf{u}}]}{\text{Var}[Y]}.$$

Describes impact of interactions of order $\mathbf{u}$ between variables with indices in $\mathbf{u}$ on the variance of Y .

We have

$$\sum_{\mathbf{u}} S_{\mathbf{u}} = \sum_{i=1}^N S_i + \sum_{1 \leq i < j \leq N} S_{ij} + \dots = 1.$$

3.5 Sobol indices: first-order

First-order indices describe **direct** impact of variables on the variance of Y , neglecting interactions:

$$S_i = \frac{\text{Var}_i [\mathbb{E}_{\sim i}[Y|\xi_i]]}{\text{Var}[Y]}.$$

$\mathbb{E}_{\sim i}$ - expectation over all variables **except** ξ_i , Var_i - variance over ξ_i .

💡 Question

What other information can we obtain from first-order Sobol indices?

💡 Impact of higher-order interactions

Since we know that $\sum S_{\mathbf{u}} = 1$, we can assess interactions between all higher-order effects as $1 - \sum S_i$.

3.6 Sobol indices: second-order and global

Incorporate pair-wise interactions:

$$S_{ij} = \frac{\text{Var}_{ij} [\mathbb{E}_{\sim(i,j)}[Y|\xi_i, \xi_j]]}{\text{Var}[Y]}.$$

We can also consider higher-order effects, and total sum of effects of varying ξ_i along with any possible interactions:

$$\hat{S}_i = S_i + \sum_j S_{ij} + \sum_{jk} S_{ijk} + \dots$$

3.7 Sobol indices: total order

The above approach for computing $\hat{S}_i$ becomes very expensive. What if we want to

- estimate overall relative impact of uncertainty/variations on ξ on Y
- do it without explicitly computing all higher-order interaction terms (number grows exponentially with number of independent parameters N)

Total-order Sobol sensitivity index:

$$S_{T,i} = \frac{\mathbb{E}_{\sim i} [\text{Var}_i[Y|\xi_{\sim i}]]}{\text{Var}[Y]} = 1 - \frac{\text{Var}_{\sim i} [\mathbb{E}_i[Y|\xi_{\sim i}]]}{\text{Var}[Y]} = 1 - S_{\sim i}.$$

Here $S_{\sim i}$ is the sum of all terms **excluding** the effect of variation in ξ_i . Unlike $\hat{S}_i$, $\sum_i S_{\sim i} \neq 1$.

4 Performing UQ/SA

How does one compute mean, variance, Sobol indices?

Computation of $\mathbb{E}[Y]$, $\text{Var}[Y]$, Sobol indices requires computation of integrals of the form

$$\int (Y - \bar{y})^k f(\xi) d\xi.$$

4.1 Monte Carlo

Sample ξ according to probability distributions, compute $Y(\xi)$, use the values to estimate integrals.

- Convergence rate: $\mathcal{O}(N_s^{-1/2})$ - independent of N
- Dependent variables: boils down to correct sampling
- Latin Hypercube, Sobol sequence: improves error, better coverage of low-probability regions

Can be unwieldy, especially if computing Y is expensive.

4.2 Polynomial Chaos Expansion

Also known as

- Generalized Polynomial Chaos
- Wiener-Askey Chaos Expansions
- Spectral Polynomial Expansions
- Stochastic Galerkin

4.3 PCE ansatz

For independently distributed variables:

$$Y(\xi) = \sum_{n=1}^{N_p} c_n P_n(\xi), \quad P_n(\xi) = \sum_{i_1+i_2+\dots+i_N=n} a_n \xi_1^{i_1} \cdot \dots \cdot \xi_N^{i_N},$$

N_p - order of expansion.

P_n are **known** polynomials of uncertain variables. c_n - unknown expansion coefficients we wish to find.

Note

If Y is dependent on spatial position $\mathbf{x}$ and/or time t , these dependencies can be incorporated into c_n .

If we have the PCE, we can integrate all the statistical expressions analytically! Or at least with the help of a computer algebra system or using exact numerical quadratures for polynomials.

4.4 Intrusive PCE

Example:

$$\frac{du}{dt} = \lambda u.$$

λ - uncertain variable dependent on random variable ξ . Expand both u and λ :

$$\lambda = \sum_n \lambda_n P_n; \quad u = \sum_m u_m P_m.$$

Plug into ODE and multiply by P_k :

$$\frac{d}{dt} \sum_m u_m P_m P_k = \sum_n \lambda_n P_n \sum_m u_m P_m P_k.$$

Question

What do we do with this?

4.5 Intrusive PCE: orthogonal polynomials

Note

If we use polynomials P_i that are orthogonal with respect to the weighting function defined by the distribution of ξ (i.e. Legendre for uniform, Hermite for normal, Laguerre for exponential, etc.), we can simplify this quite a bit!

4.6 Orthogonal polynomial expansion

$$\frac{d}{dt} \sum_m u_m P_m P_k = \sum_n \lambda_n P_n \sum_m u_m P_m.$$

By integrating over ξ and using orthogonality of P_k we get

$$\frac{du_k}{dt} = \sum_n \sum_m \lambda_n u_m C_{mnk},$$

where

$$C_{mnk} = \frac{\int P_m P_n P_k d\xi}{\int P_k^2 d\xi} =: \frac{\langle P_m P_n P_k \rangle}{\langle P_k^2 \rangle}.$$

Coefficients C_{mnk} are independent of problem solution, depend only on the polynomial basis - can be precomputed.

4.7 Intrusive PCE

Achieves good convergence rates, but requires

- derivation of equations for the expansion coefficients
- modifying the source code of the simulation software
- solution of additional PDEs (with all associated issues - stability, convergence, etc.)

4.8 Non-intrusive quadrature-based PCE

Take ansatz (with orthogonal polynomials)

$$Y(\xi) = \sum_{n=1}^{N_p} c_n P_n(\xi)$$

and use orthogonality property:

$$c_n = \frac{\langle Y P_n \rangle}{\langle P_n^2 \rangle}.$$

Question

How do we compute the inner products $\langle Y P_n \rangle$?

Answer

We use a numerical quadrature rule!

4.9 Non-intrusive quadrature-based PCE

In non-intrusive quadrature-based PCE, we estimate $\langle Y P_n \rangle$ as

$$\langle Y P_n \rangle \approx \sum_j \omega_j Y(\xi_j) P_n(\xi_j).$$

Each value $Y(\xi_j)$ is obtained by running the simulation for the parameter ξ - value taken from quadrature point.

Can use quadrature rule of choice, but:

- Quadrature related to distribution of ξ will perform better (reduce aliasing)
- Clenshaw-Curtis allows for re-use of values
- High-dimensional parameter spaces ($N \approx 10$ or larger) require sparse grids to due curse of dimensionality
 - Sparse grids may lead to unstable results (negative quadrature weights are possible)

4.10 Least-squares based PCE

Alternate approach:

$$\sum_{n=1}^{N_p} c_n(\mathbf{x}) P_n(\xi_i) = Y_i, \quad i = 1, \dots, N_{sample} \Leftrightarrow$$
$$\begin{pmatrix} P_1(\xi_1) & \dots & P_{N_p}(\xi_1) \\ \vdots & \ddots & \vdots \\ P_1(\xi_{N_{sample}}) & \dots & P_{N_p}(\xi_{N_{sample}}) \end{pmatrix} \begin{pmatrix} c_1 \\ \vdots \\ c_{N_p} \end{pmatrix} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_{N_{sample}} \end{pmatrix}.$$

Set $N_{sample} > N_p$ and solve in least-squares sense.

Values ξ_i obtained by sampling from distribution. Can use adaptive approaches for better parameter space exploration (e.g. 1, 2).

4.11 PCE summary

- Intrusive PCE: solve ODEs/PDEs for expansion coefficients
- Non-intrusive PCE with quadrature: obtain values from solver for by performing simulation for ξ_i obtained from quadrature rule, estimate quadrature
- Non-intrusive PCE with sampling: sample ξ_i from distribution, construct and solve linear system

5 Inverse parameter estimation

PCE gives use a **reduced-order model**: simple evaluation of polynomials to obtain Y for a given value of ξ .

If we obtain better data for Y (for example, from experiments), we can use this to reduce uncertainty in ξ .

i Note

Bayes' theorem:

$$P(\xi|Y) = \frac{P(Y|\xi)P(\xi)}{P(Y)}, \quad P(Y) = \int P(Y|\xi)P(\xi)d\xi.$$

Use Markov Chain Monte-Carlo sample $P(\xi|Y)$ with acceptance probability $\alpha = \min \left[1, P(Y|\xi^*)P(\xi^*) / (P(Y|\xi^{(j)})P(\xi^{(j)})) \right]$. Accept ξ^* as ξ^{j+1} with probability α .

6 Parameterization

Example of how physics/data-driven parameterizations can lead to lower parameter numbers and reduce inter-parameter dependencies.

6.1 VT relaxation

In high-enthalpy flows, vibrational energy is not in equilibrium with translational energy. A separate equation for vibrational energy of the molecular species c is solved:

$$\frac{d(\rho_c E_v^c)}{dt} = -\nabla \cdot \mathbf{q}_{v,c} + R_c^{vibr} - \rho_c E_v^c \nabla \cdot \mathbf{v}.$$

The relaxation is oftentimes modeled via the Landau-Teller equation:

$$R_c^{vibr,VT} = \rho_c \frac{E_v^c(T) - E_v^c(T, T_v^c)}{\tau_c^{VT}}.$$

From experimental data Millikan and White obtained a fit:

$$p\tau_{cd}^{VT} = \exp(A_{cd}(T^{-1/3} - B_{cd}) - 18.42)$$

6.2 VT relaxation time parameterization

We cannot vary A_{cd} , B_{cd} independently, as that may lead to very low or large values of τ_{cd}^{VT} .

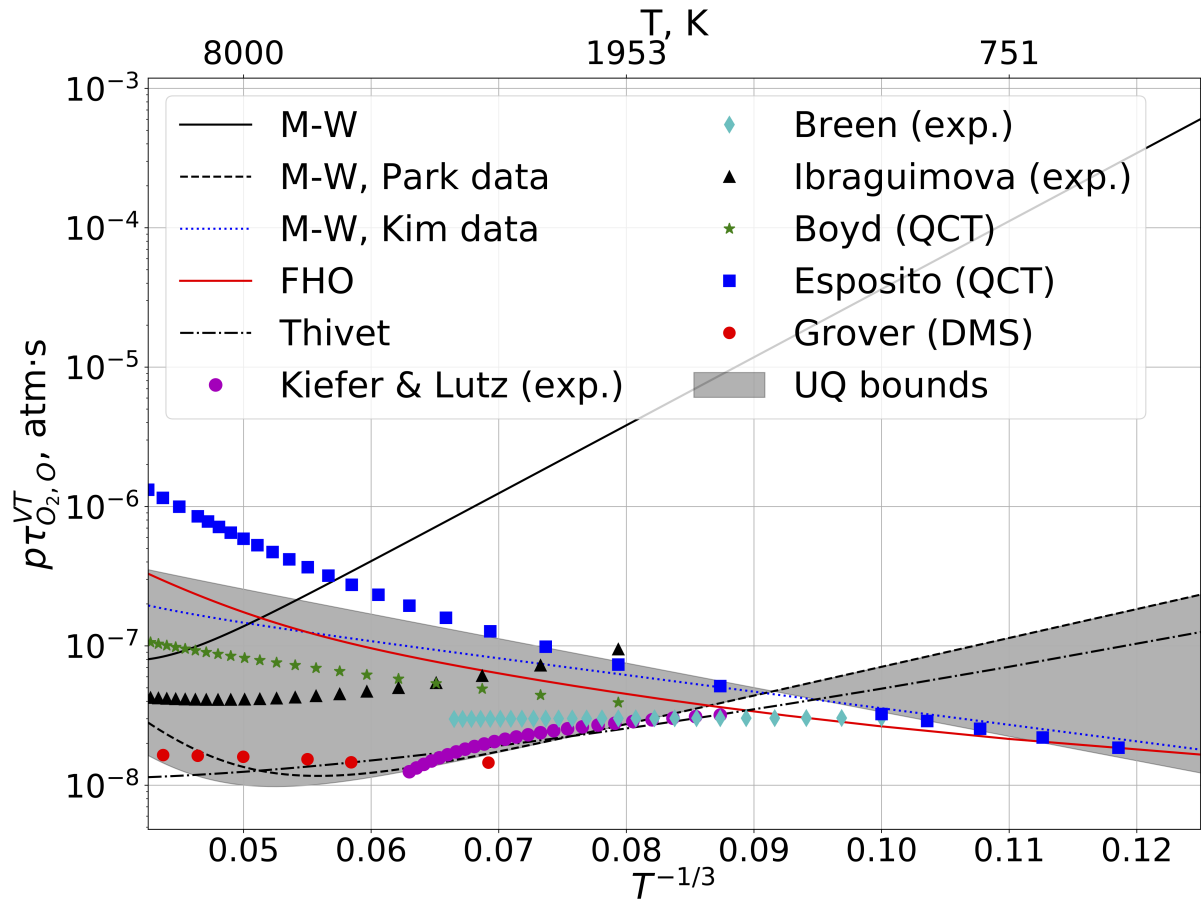


Figure 1: Data for O₂-O VT relaxation

Based on analysis of data, introduce $\alpha_{cd}^{VT} \in [0, 1]$:

$$A_{cd} = A_{cd}^{(0)} + A_{cd}^{(1)} \alpha_{VT,cd}^{VT}, \quad B_{cd} = \frac{A_{cd}^{(0)} B_{cd}^{(0)} + A_{cd}^{(1)} B_{cd}^{(1)} \alpha_{cd}^{VT}}{A_{cd}^{(0)} + A_{cd}^{(1)} \alpha_{cd}^{VT}}.$$

7 Software

- [Chaospy](#)
 - [EasyVVUQ](#): built on top of Chaospy
- [UQTK](#)
- [PolyChaos.jl](#)

8 Questions

- What are the different statistical properties used in UQ? What is the difference between global and local sensitivity analysis?
- What is the decomposition of variance, and how can it be used to define sensitivity indices?
- What is PCE? How do the different versions (intrusive/non-intrusive with quadratures/non-intrusive with least squares) work?

9 References

- R. C. Smith, Uncertainty Quantification: Theory, Implementation, and Applications. Society for Industrial and Applied Mathematics, 2013